PROJECT PROPOSAL

Campus Equipment & Laboratory Reservation Management System

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The Problem:

Schools and laboratories often have shared equipment and spaces that many students and faculty members need to use. When reservations are done manually, problems such as **double-booking, equipment loss, and delays** can occur.

The proposed system will provide a simple Python-based reservation system where users can check available equipment, make reservations, and manage equipment checkout and check-in.

The Objective of our Project

The project aims to:

1. Create a simple system for reserving laboratory equipment and spaces.
2. Prevent **double-booking** by checking for scheduling conflicts.
3. Allow users to view available equipment.
4. Track equipment checkout and check-in.
5. Store and display reservation records.
6. Provide different access options for students, faculty members, and laboratory custodians.
7. Validate user inputs to reduce errors.

Planned Features of our Project

- **Student reservation** – Students can reserve available equipment.
- **Faculty reservation** – Faculty members can make reservations.
- **Lab custodian management** – Custodians can manage equipment and reservations.
- **View available equipment** – Users can see equipment that can be reserved.
- **Conflict checking** – The system checks if an equipment/time slot is already reserved.
- **Checkout and check-in tracking** – Tracks whether equipment has been borrowed or returned.
- **Reservation records** – Stores information about reservations.
- **Input validation** – Checks whether the user's input is valid.
- **Role-based access** – Different options are available depending on the user's role.

Planned Inputs and Outputs:

Inputs

The system may ask the user for:

- User role
  - Student
  - Faculty
  - Lab Custodian
- User name
- Equipment to reserve
- Reservation date
- Time slot
- Checkout/check-in information
- Menu choice

Outputs

The system will display:

- Available equipment
- Reservation confirmation
- Reservation details
- Reservation status
- Equipment checkout/check-in status
- Error messages for invalid inputs
- Scheduling conflict messages

Example

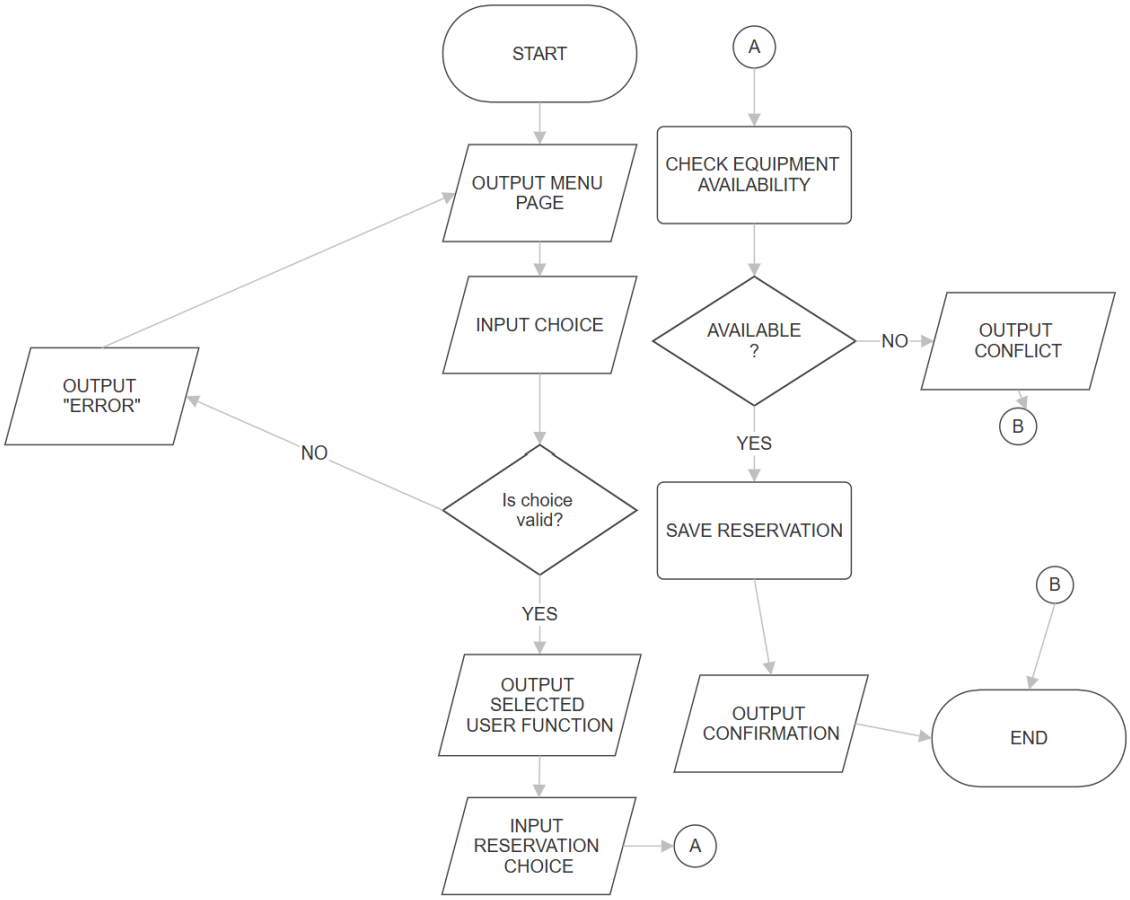
Input:

```
Role: Student
Name: John
Equipment: Microscope
Date: 2026-09-01
Time: 10:00 AM
```

Output:

```
Checking availability...
Reservation successful!
Equipment: Microscope
Reserved by: John
Date: 2026-09-01
Time: 10:00 AM
Status: Reserved
```

Logic Plan / Flowchart:



Contributors

- **[FAUSTINO]** — Reservation system and conflict-prevention logic
- **[ALBOS]** — Equipment checkout and check-in tracking
- **[ABARCA]** — Role-based access, testing, and documentation